



Improving Safety During Repair Management of the 994K

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

In most mining sites around the world, the safety policy, as well as compliance issues, push mining companies to set up continuous improvement (CI) projects to make the maintenance process better, protecting the lives of technicians during the execution.

In 2020, the first 994K loaders began operating at the S11D mine in Brazil, as part of the customer’s effort to increase iron ore production and increase asset flexibility. The 994K Wheel Loader has a steering frame lock link that maintains alignment between both frames. Sotreq has identified an improved process in handling this link.



Figure 1 – 994K Wheel Loader

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2.0 Best Practice Description

The 994K Wheel Loader steering frame lock link is used during maintenance to avoid crush risk and increase the security of the technicians. The steering lock link has two pins, one in each end.

The image below, shows the details of the steering frame lock link:



Figure 2 – 994K steering frame lock link

To improve the mechanics' safety, a handling device was welded onto the top of each steering lock pin and on two locations of the steering lock link. These handling devices will reduce the risk of serious injury to personnel.

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3.0 Implementation Steps

SOTREQ removed both pins PN 124-3820 and the link 391-2310 from the machine. A steel bar of 3/8” was welded on the top of both pins and at two locations on the link to create handles. When welding, use pre-heat and low hydrogen wire to reduce the risk of cracking. Inspect the heat affected zone of pins and link for surface cracks up to 48 hours after welding.



Figure 3 – (1) Pins PN 124-3820 / (2) Drawing / (3) After Best Practice – Source: Sotreq/Wender Batista

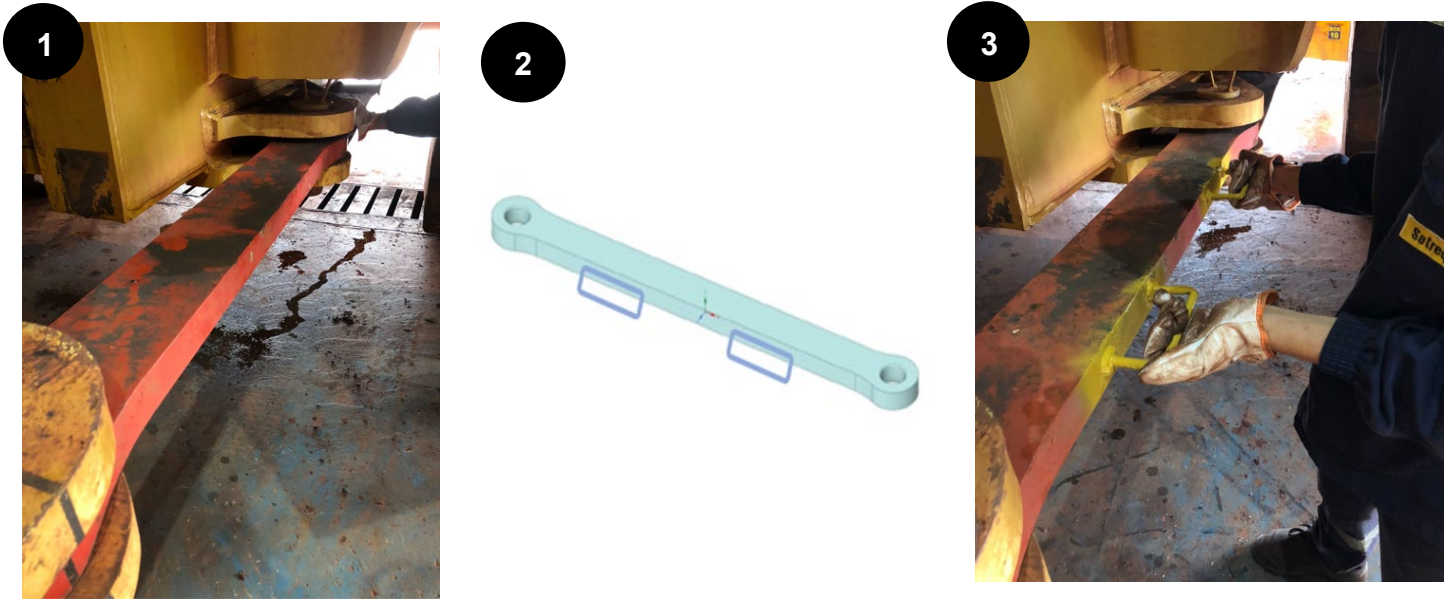


Figure 4 – (1) Link PN 391-2310 / (2) Drawing / (3) After Best Practice – Source: Sotreq/Wender Batista

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4.0 Benefits

- Safe and efficient removal and install of the steering frame lock link;
- Less risk to technicians during maintenance and repair processes;
- Improved ergonomics.

5.0 Resources Required

Item	Cost
Material, ASTM A36 Steel Bar	\$10
Labor, Welder	\$50

6.0 Supporting Attachments / References

Drawings are attached

7.0 Related Best Practices

- 793D Final Drive Planetary Group Lifting Tool
- Exhaust Manifold Lifting Tool – C175 Engines
- R&I Tool for 797 Steering Cylinders
- Tool for R&I of the 797F High Pressure Fuel Pump

8.0 Acknowledgements

SOTREQ would like to express our admiration for the professionals involved in improving the maintenance and repair process.

SOTREQ thanks the technical contribution of **Doug Friedrich** of **Caterpillar Inc**; Thanks Doug.

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